MARIOS GLYTSOS

Brooklyn, New York
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RESEARCH INTERESTS

Audio Representation Learning · Generative Audio · Multimodal Perception · Music Information Retrieval · Digital Signal Processing

EDUCATION

New York University **New York, NY**
Ph.D. in Music Technology *Sep. 2025 – Exp. 2030*

- **Advisor:** Brian McFee | **Current GPA:** 4.0/4.0
- **Interests:** Audio Representation Learning, Generative Audio, Multimodal Machine Perception, Spatial Audio
- **Current Research:** Investigating source-specific information recovery in pretrained audio representations through generative reconstruction.

National Technical University of Athens **Athens, Greece**
Joint BSc & MEng in Electrical and Computer Engineering *Sep. 2017 – May 2024*

- **GPA:** 9.25/10.
- **Thesis:** *Music Source Separation on Classical Guitar Duets.*

RESEARCH EXPERIENCE

Athena Research Center **Athens, Greece**
Researcher — euROBIN / SynthPose *Sep. 2024 – Aug. 2025*

- Developed pose estimation methods for agricultural robotics by generating realistic synthetic datasets with generative models. Leveraged lattice deformations to model intra-class shape variations.
- Contributed to deep learning pipelines for object detection and pose estimation, integrating real and synthetic data.

B' Radiology Laboratory, NKUA Medical School **Athens, Greece**
Research Assistant *Feb. 2024 – Aug. 2024*

- Developed and evaluated AI algorithms for estimating treatment plan recomputation needs in image-guided radiotherapy.

DEEPLAB, NTUA **Athens, Greece**
Research Intern *Jul. 2022 – Sep. 2022*

- Conducted research on visual common-sense reasoning and deep learning for visual understanding.

PUBLICATIONS

– **M. Glytsos** and B. McFee, "How Much Audio Is Left in an Embedding? An Inversion Audit of Audio Encoders," *Under submission to IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2027.

– N. Chatzis, A. Tsinouka, K. Papadimitriou, N. Efthymiou, **M. Glytsos**, G. Retsinas, P. Oikonomou, G. Potamianos, P. Maragos, and P. P. Filntisis, "Mind the Shape Gap: A Benchmark and Baseline for Deformation-Aware 6D Pose Estimation of Agricultural Produce," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

– **M. Glytsos**, P. Filntisis, G. Retsinas, and P. Maragos, "Category-Level 6D Object Pose Estimation in Agricultural Settings Using a Lattice-Deformation Framework and Diffusion-Augmented Synthetic Data," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

– A. Garoufis, **M. Glytsos**, E. Chourdaki, P. Filntisis, and P. Maragos, "Power in Unity: Combining In-Domain and Out-of-Domain Pre-Training Strategies for EEG-Based Person Identification," *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2025.

– **M. Glytsos**, A. Garoufis, N. Zlatintsi, and P. Maragos, "Classical Guitar Duet Separation Using GuitarDuets – A Dataset of Real and Synthesized Guitar Recordings," *International Society for Music Information Retrieval Conference (ISMIR)*, 2024.

TEACHING & ACADEMIC SERVICE

Teaching Assistant — Fundamentals of Digital Signal Theory (MPATE-GE 2599) Fall 2026

- Supported course instruction covering sampling, aliasing, convolution, discrete-time systems, and Fourier analysis; managed learning platform, grading, and student advising.

Reviewer — International Society for Music Information Retrieval Conference (ISMIR) 2025, 2026

TECHNICAL LEADERSHIP & ENTREPRENEURSHIP

CAIUS **Metsovo, Greece**
Co-Founder & AI Lead *May 2022 – Aug 2025*

- Co-founded and lead AI development for a geospatial-AI system mapping forest trails and sustainability solutions using satellite data.
- Secured €125k in funding (**ESA BIC**, **CASSINI Accelerator**) and recognized by **Forbes Greece 30 Under 30**.

SELECTED HONORS & COMPETITIONS

- **2nd & 3rd Place**, Cadenza Challenge, Task 2, 2025.
- **3rd Place**, ICASSP 2025 Grand Challenge on EEG-Music Emotion Recognition, NTUA-IRAL Team.
- **Forbes Greece 30 Under 30**, 2024.
- **1st Place**, European CASSINI Hackathon, 2022.

TECHNICAL SKILLS

Programming: Python, C++, C, Java, JavaScript, SQL
Machine Learning: PyTorch, TensorFlow, NumPy, SciPy, scikit-learn
Research & Systems: Linux, Git, AWS, FastAPI, React, Multi-GPU/Distributed Training
Audio & Music: JUCE, Reaper, Logic Pro, FL Studio, Cubase, MuseScore, iZotope RX